NLP Phase 1 Presentation

Optimal Trajectory for LLM based Gaming-Domain Synthesis

Document Representation, Uni/Multilingual Query Processing,
Optimal Retieval and Evidence-Grounded Synthesis

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Problem Statement

Answering gaming questions from specialized, changing English documentation

Gaming questions need accurate evidence about mechanics, names, technical issues and version-specific changes. That evidence may be spread across documents, phrased differently from the question, or queried in a language or script other than the English corpus.

Why the Gaming Domain Is Hard

- **Constant change:** Patches make older answers wrong.
- **Dense jargon:** "Bene Gesserit", "FG3".
- **Chatty questions:** The real ask is buried.
- **Look-alike documents:** Near-duplicate pages compete.

Why Hindi and Romanized Hindi Are Harder

- **Script mismatch:** Hindi queries, English documents.
- **No standard spelling:** "hai", "haye", "hain".
- **Names break:** "PUBG" → "pabji"; "lasguns" → "lasagna".

Research problem: How can a gaming-domain retrieval and RAG system represent documents, process queries, retrieve relevant evidence and synthesize grounded answers reliably?

Scope: Study each stage separately; evaluate the integrated pipeline if no phase yields a strong standalone contribution.

Why This Problem Matters

Find relevant evidence before generating an answer

- **Ground answers in evidence:** Retrieve source documents that support generated responses.
- **Access specialized knowledge:** Retrieve game mechanics, patch information, mods and technical solutions.
- **Handle changing information:** Retrieve documentation for specific game versions and updates.
- **Bridge vocabulary gaps:** Match queries to documents describing the same concept differently.
- **Enable cross-lingual access:** Investigate whether Hindi and Romanized Hindi queries retrieve relevant English documents.

Example 1

"How do I harvest Spice during a Spice Blow without risking an overdose?"

Retrieve relevant mechanics and risk guidance.

Example 2

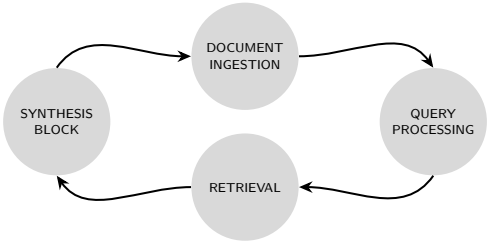
"How do I deal with the giant worm in that desert survival game?"

Retrieve relevant guides despite an underspecified query.

One Pipeline, Four Research Opportunities

Four connected phases, each with its own research question and evaluation

Phase	Research question
1. Documents	How should gaming documents be represented and organized to support retrieval?
2. Queries	How should English, Hindi and Romanized Hindi queries be represented or transformed?
3. Retrieval	How can the system retrieve and rank the most relevant gaming evidence?
4. Synthesis	How can retrieved evidence become concise, complete, grounded answers?



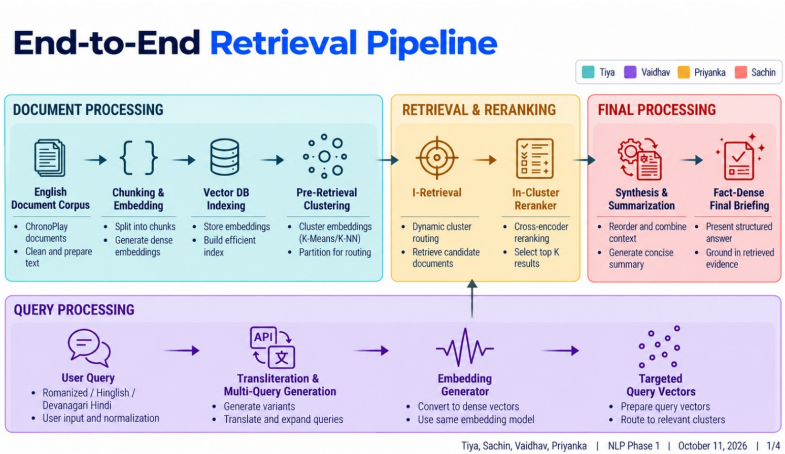
Standalone option: A phase with a defensible result can become the main project.

Integrated option: Otherwise, the phases form one evaluated end-to-end system.

Arrows show the integrated system's information flow, not a requirement that all four phases succeed.

Team Responsibilities and System Overview

Proposed integrated system and allocation of work



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Reading the figure: It shows the proposed integrated system and who leads each block. Work allocation does not decide whether the final contribution must span every phase.

ChronoPlay-QA: Dataset and Motivation

A gaming-domain benchmark for retrieval and RAG

4,192	**232**	**3**
Corpus documents	Original query–answer pairs	Games covered

Why This Benchmark?

- Gaming RAG is new: "there are currently no RAG benchmarks for the gaming domain" (He et al., ICLR 2026).
- Specialized terminology: game titles, characters, items and error codes.
- Reference answers that can be mapped to corpus documents for relevance judgments.

Two Ways to Describe a Question

- **Content area** (what it is about): gameplay mechanics, technical support, game updates.
- **Task type** (what it asks the system to do): comparative, extraction-based, multi-hop reasoning.

He et al. (2026), *ChronoPlay: A Framework for Modeling Dual Dynamics and Authenticity in Game RAG Benchmarks.* Paper

Dataset Statistics and Evaluation Risks

The corpus is imbalanced, and relevance judgments are sparse

Statistic	Corpus	Queries	Answers
Items	4,192	223	223
Word tokens	1,006,493	5,867	11,384
Vocabulary	23,743	1,469	2,611
Avg. tokens/item	240.1	26.3	51.0

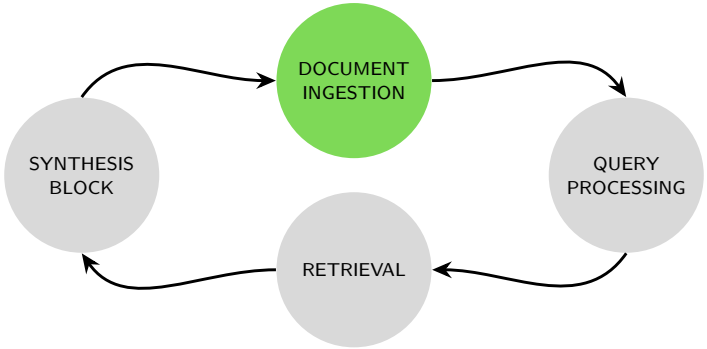
Game	Documents	Queries
Dune: Awakening	3,512 (83.8%)	49
Dying Light 2	513 (12.2%)	85
PUBG Mobile	167 (4.0%)	89

Coverage and Labels

- **223 of 232 queries usable:** 9 excluded because their reference passages could not be reliably matched to any corpus document.
- **Labels are derived, not given:** the dataset ships reference text only. We mapped it to document IDs via exact titles and verbatim text fragments.
- **Sparse:** only 112 distinct documents (2.7% of the corpus) are ever relevant; median 1 per query, max 13. A single rank position can move a score.

The raw `qa_pairs` table has 232 rows with these columns: `game_name`, `question`, `answer`, `references`, `retrieved_docs`, `question_topic`, `task_type`, `time`, `entities`

Phase 1: Document Ingestion and Clustering



Research question: How should gaming documents be represented and organized to support effective retrieval?

Phase 1: Research Question and Scope

Building meaningful document representations

Question: How do preprocessing, text representation, embedding adaptation and clustering affect the organization and downstream retrievability of gaming documents?

1. Preprocessing

Compare pipelines A, B and C: how does text cleaning affect retrieval and representations?

2. Representation

Compare text fields, TF-IDF versus dense embeddings, encoder models and document granularity.

3. Representation Learning

Test whether contrastive adaptation improves document embeddings.

4. Clustering

Compare clustering baselines and evaluate a precisely defined hierarchical variant.

Experiment 1: Preprocessing Pipelines

How much should we transform the original documents?

Pipeline A — Linguistic

- Hybrid tokenization.
- POS-aware lemmatization.
- Preserve selected names, numbers and domain terms.

Pipeline B — Aggressive

- Lowercase all text.
- Remove NLTK stop words.
- Apply Snowball stemming.

Pipeline C — Conservative

- Normalize Unicode and whitespace.
- Decode HTML entities and remove markup.
- Preserve wording, casing and terminology.

Rationale: Tests whether linguistic transformation helps, harms or leaves retrieval unchanged. Corpus, queries, encoder and evaluation stay fixed.

Experiment 1: Preliminary Results

Preprocessing pipelines on a ChronoPlay-QA

Pilot setup: 200 documents, 12 queries, the same `all-MiniLM-L6-v2` encoder for pipelines A, B and C.

Primary Retrieval Results

Metric	A	B	C
MRR	0.7389	0.5868	**0.8958**
Recall@5	1.0000	0.8333	**1.0000**
Hit@5	1.0000	0.8333	**1.0000**

- **Best ranking:** C ranks the first relevant result highest on average.
- **Top-five coverage:** A and C reach 1.0000; B reaches 0.8333.

Secondary Representation Findings

	Mean cosine	Cluster sizes ($k = 4$)
A	0.3066	99, 56, 9, 36
B	0.4027	130, 37, 9, 24
C	0.2239	39, 92, 42, 27

B has the highest mean cosine and one 130-document cluster; C is less concentrated. Neither establishes better clustering.

Status: With 12 queries, C being best is not established for the full benchmark; C is carried forward as the working baseline, with raw text and stable document IDs preserved.

Experiment 2: Text Fields and Representation

Each comparison tests one representation decision

Decision	Conditions	Why compare them
Text field [1]	Title only; contents only; title + contents	Concise topic signal versus detail versus both.
Sparse vs. dense [2, 3]	TF-IDF; dense embeddings	Exact term overlap versus meaning beyond word matches.
Encoder [4–6]	`all-MiniLM-L6-v2`; `all-mpnet-base-v2`; `intfloat/e5-base-v2`	Efficient baseline; higher capacity; retrieval-trained embeddings.
Granularity [7]	Whole record; chunks	Chunk only if a token-length audit shows truncation; preserve chunk-to-document ID mapping.

References

[1] Trotman, 2005 — *Choosing Document Structure Weights.*
[2] Karpukhin et al., EMNLP 2020 — *Dense Passage Retrieval for Open-Domain Question Answering.*
[3] Subakti et al., 2022 — *The Performance of BERT as Data Representation of Text Clustering.*
[4] Reimers & Gurevych, EMNLP-IJCNLP 2019 — *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks.*
[5] Muennighoff et al., EACL 2023 — *MTEB: Massive Text Embedding Benchmark.*
[6] Wang et al., 2022 — *Text Embeddings by Weakly-Supervised Contrastive Pre-training.*
[7] Chen et al., EMNLP 2024 — *Dense X Retrieval: What Retrieval Granularity Should We Use?*

Control: Keep corpus, queries, preprocessing and evaluation protocol fixed for each comparison. Balance quality against encoding cost, memory and truncation.

Experiment 3: Adaptive Representation Learning

Can contrastive learning improve document representations and retrieval?

1. Pretrained Baseline

Use the selected pretrained encoder without additional training as the reference for evaluating representation adaptation.

2. Unsupervised Contrastive Learning

SimCSE-style adaptation: Create two representations of the same sentence using different dropout masks. Contrast them as a positive pair against other sentences in the batch.

Purpose: Test whether adapting representations using unlabeled gaming text improves semantic structure and retrieval without requiring relevance labels.

3. Supervised Contrastive Adaptation

Training signal: Queries act as anchors, relevant documents as positives, and carefully selected non-relevant documents as negatives.

Purpose: Test whether task-specific supervision aligns query and document representations more effectively than unsupervised adaptation.

Evaluation: Compare the pretrained, unsupervised and supervised models using retrieval metrics and, where meaningful, cluster quality and stability.

Gao, Yao & Chen (2021), *SimCSE: Simple Contrastive Learning of Sentence Embeddings*. Paper Karpukhin et al. (2020), *Dense Passage Retrieval for Open-Domain Question Answering*. Paper

Experiment 4: Comparing Clustering Algorithms

Which document organization supports retrieval best?

Partitioning and Hierarchy

K-Means

- Scalable baseline with a predefined number of clusters.

Agglomerative Clustering

- Builds nested groupings at different levels.

RAPTOR (Sarthi et al., 2024)

- Recursively clusters and summarizes text to build a hierarchy for retrieval at multiple levels.

Evaluation Criteria

- **Cluster quality:** Silhouette and stability.
- **Structure:** Cluster sizes, noise and interpretability.
- **Question-type analysis:** Compare downstream retrieval for extraction, comparative and multi-hop queries where labels are available.

Density-Based Clustering

HDBSCAN

- Finds density-based groups and identifies potential noise.

BERTopic

- Embedding-based topic groupings and topic representations.

Carrot^2

- Search-result clustering using methods such as Lingo or STC.

Sarthi et al. (2024), *RAPTOR: Recursive Abstractive Processing for Tree-Organized Retrieval*. Paper Campello et al. (2013), *Hierarchical Density Estimates for Data Clustering, Visualization, and Outlier Detection*. Paper
Grootendorst (2022), *BERTopic*. Paper Rousseeuw (1987), *Silhouettes: A Graphical Aid to the Interpretation and Validation of Cluster Analysis*. Paper
He et al. (2026), *ChronoPlay*. Paper

Phase 1: Expected Contribution and Decision Criteria

What would make Phase 1 a standalone result?

Standalone Contribution If...

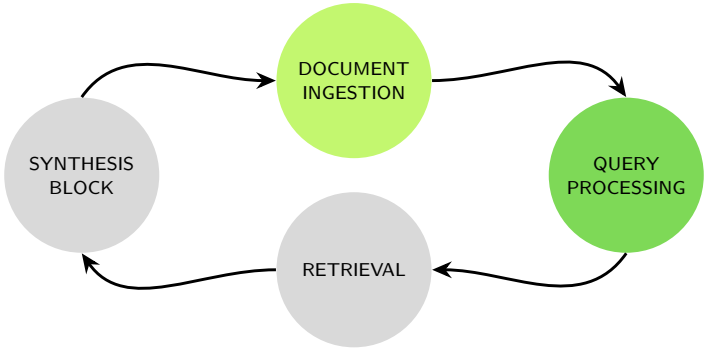
- A representation or clustering choice gives a **repeatable, measurable** gain,
- supported by **controlled ablations** (one change at a time),
- and confirmed by **downstream retrieval** on the full benchmark, not only cluster statistics.

Otherwise

- Carry the best-supported representation forward as a fixed component.
- Use clusters in the integrated system only if they beat flat retrieval.

Handoff to the system: Document representations and, if justified, clusters for downstream retrieval.

Phase 2: Query Processing



Research question: How should English, Hindi and Romanized Hindi queries be represented or transformed for retrieval?

Shared Query Cleaning and Preprocessing

Shared methodology for Plans A and B, not a separate plan

Is Query Cleaning Necessary?

Condition	Processing
C0 — Original	Query as supplied.
C1 — Formatting	Normalize Unicode, whitespace and markup.
C2 — Conservative	Reliable entity aliases and identifiable typo correction.

Avoid by Default

Technique	Risk
POS filtering	Drops context and modifiers.
NER	Does not by itself improve retrieval.
Lemmatization	Alters word forms before encoding.
Stemming	Distorts names and game terms.
Stop-word removal	Removes negation and parts of names.

Principle: Preserve the original query unless a transformation has a clear purpose and a measurable benefit.

Typo robustness: Zhuang & Zuccon (2022), *CharacterBERT and Self-Teaching for Improving the Robustness of Dense Retrievers on Queries with Typos*. Paper
Retrieval context: Karpukhin et al. (2020), *Dense Passage Retrieval for Open-Domain QA* Paper; it does not show that each listed operation harms retrieval.

Plan A: Improving English Query Representations

Can query expansion and query–document alignment improve retrieval?

Experimental Approaches

1. **Original query:** Baseline using the Phase 1-selected encoder.
2. **Related-term expansion:** Add related gaming terms and synonyms.
3. **Query2doc (Wang et al., 2023):** An LLM generates a pseudo-document describing the query. Append it to the original query to test whether richer context improves retrieval.
4. **LLM-generated variations + fine-tuning:** Generate intent-preserving query variants and use validated synthetic query–document pairs to adapt the encoder.
5. **Contrastive fine-tuning:** Align queries with verified relevant documents using a contrastive learning objective.
6. **Query decomposition:** Split multi-part questions into sub-queries and combine the retrieved evidence.

Evaluation Metrics

- **nDCG@10:** Top-ranked relevance.
- **MRR@10:** Rank of the first relevant document.
- **Recall@5 / Recall@10:** Relevant-document coverage.

Expected Outcome

Identify which approaches improve query–document alignment and retrieval

Wang, Yang & Wei (2023), *Query2doc*. Paper Dai et al. (2023), *Promptagator*. Paper Gao et al. (2021), *SimCSE*. Paper
Ammann et al. (2025), *Question Decomposition for RAG*. Paper Zheng et al. (2023), *Take a Step Back*. Paper

Plan B: Does Language or Script Affect Retrieval?

Comparing retrieval across languages, scripts and encoder types

B0 — English	B1 — Hindi	B2 — Romanized Hindi
Original English query.	Translation in Devanagari.	Validated Hindi transliterated into Latin script.

Procedure

1. Translate English queries into Hindi using IndicTrans2.
2. Transliterate validated Hindi using IndicXlit.
3. Preserve game names, item names, numbers and versions.
4. Retrieve all query forms against the same English corpus.

Key limitation: Translated-then-transliterated Hindi is not necessarily naturally typed Hinglish.

Encoders to Compare

- **English encoder:** Best-performing encoder selected from Phase 1, based on development results.
- **Multilingual encoder:** BGE-M3 and Multilingual E5-large, evaluated on all three query forms.

Expected Outcomes

- Measure language and script-related retrieval gaps.
- Identify which encoder is more robust across query forms.

Gala et al. (2023), *IndicTrans2*. Paper Madhani et al. (2023), *Aksharantar: Open Indic-language Transliteration Datasets and Models*. Paper Chen et al. (2024), *BGE-M3*. Paper

Plan B: Reducing the Cross-Lingual Retrieval Gap

Can query reformulation and encoder adaptation improve Romanized Hindi retrieval?

Experiments to Compare

1. **Direct multilingual retrieval:** Above best model
2. **Translate-to-English retrieval:** Translate queries into English before retrieving from the English gaming corpus.
3. **LLM-generated query variations:** Generate multiple intent-preserving Romanized Hindi variants, retrieve for each, and fuse their ranked results.
4. **Contrastive encoder fine-tuning:** Generate and validate Romanized query variants, pair them with relevant English documents, and fine-tune the multilingual encoder to improve query–document alignment.

Expected Outcome

Determine whether query normalization, multi-query retrieval, or contrastive adaptation reduces the gap between Romanized Hindi and English retrieval.

Evaluation Metrics

- nDCG@10
- MRR@10
- Recall@5 / Recall@10

Chen et al. (2024), *BGE-M3*. Paper Wang et al. (2024), *Multilingual E5*. Paper
Madhani et al. (2023), *Aksharantar*. Paper Li et al. (2024), *DMQR-RAG: Diverse Multi-Query Rewriting for RAG*. Paper

Phase 2: Research Gap and Expected Contribution

Two different questions: query expansion (A) and cross-lingual retrieval (B)

Plan A — English Query Representation

Gap: Query2doc reports gains on general web-search benchmarks (MS MARCO, TREC DL), not on gaming queries dense with item names and codes, where generated text may invent or alter entities.

Standalone contribution: Reliable evidence that a query-representation strategy improves English gaming retrieval over the fixed baseline.

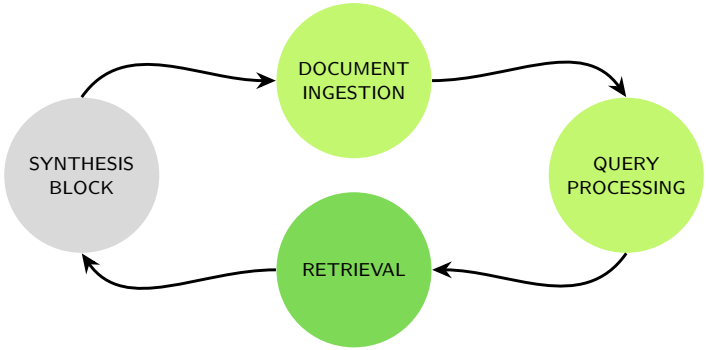
Plan B — Language and Script

Gap: Separate *language* effects (Hindi vs. English) from *script* effects (Devanagari vs. Latin), and test whether methods that work for translated Hindi also work for Romanized queries. Existing Hindi–English IR is related work, not an exact Romanized-Hindi baseline.

Standalone contribution: A measured language/script gap and a method that reduces it under controlled conditions.

Otherwise: If neither plan shows a defensible gain, the integrated system uses the strongest supported query path.

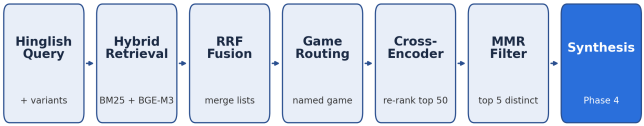
Phase 3: Retrieval and Re-ranking



Research question: How can the system retrieve and rank the most relevant gaming evidence, for English and Hindi/Romanized queries alike?

Phase 3: Pipeline and Baseline Paper

Same retrieval core for both query paths; documents are always English



Stage	Method (reference)
Language check	Devanagari, or $\geq$0.6 of words in corpus vocab. $\Rightarrow$ English
Lexical + dense	BM25 + BGE-M3 dense (Chen et al., 2024)
Rank fusion	Reciprocal Rank Fusion, $k = 60$ (Cormack et al., 2009)
Game routing	Filter to the named game (Kulkarni & Callan, 2015)
Re-ranking	Cross-encoder, top 50 (Nogueira & Cho, 2019)
Hand-off	MMR, $\lambda = 0.7$: 5 distinct docs (Carbonell & Goldstein, 1998)

Primary Reference: BGE-M3

One multilingual model (100+ languages) for dense retrieval over long inputs, so Hindi queries and English documents share one vector space.

Boundary: BGE-M3 motivates the multilingual retriever; it does not validate hybrid fusion, routing or re-ranking. Each is a separate method, tested by ablation.

Chen et al. (2024), *M3-Embedding*, Findings of ACL 2024 Paper Cormack et al. (2009) Paper Kulkarni & Callan (2015) Paper Nogueira & Cho (2019) Paper Carbonell & Goldstein (1998) Paper

Phase 3: Experiments and Preliminary Results

nDCG@10 on all 223 usable queries; each row adds one component

Stage	English	Romanized	Hindi
BM25	0.636	0.369	0.026
BGE-M3 dense	0.685	0.234	0.558
Hybrid (RRF)	0.777	0.373	0.198
+ game filter[a]	0.772	0.373	0.208
+ re-ranker	**0.823**	0.473	0.710
Hinglish path: all variants fused + re-ranked			**0.805**

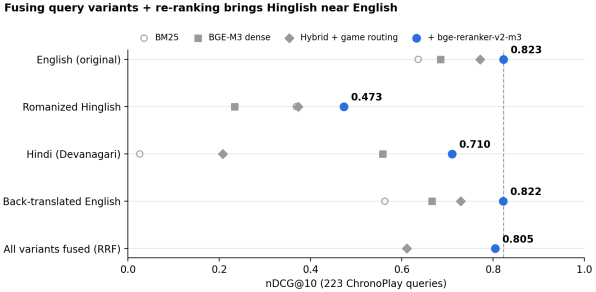
[a] Main run filtered before fusion. Filtering after fusion, computed offline from saved runs: 0.780 / 0.387 / 0.231; re-ranker re-run on this setting pending. Strict 202-query subset: 0.829 / 0.484 / 0.709; Hinglish path 0.808.

- **English:** fusion is the largest gain (+0.141 over BM25, $p < 0.001$); final Hit@5 0.897.
- **Re-ranking:** +0.051 / +0.100 / +0.501 over the filtered hybrid (all $p < 0.01$).
- **Script gap:** on Romanized queries BGE-M3 falls to 0.234, below BM25 (cf. Chari et al., 2025).
- **Naive fusion fails on Devanagari:** BM25 finds almost nothing (0.026), pulling hybrid to 0.198.
- **Hinglish path:** 0.805 vs. 0.823 English ($p = 0.13$); Hit@5 0.892.
- **Language check:** 100% correct on English, Romanized and Hindi; 99.6% on back-translated English.

Caveat: Hindi and Romanized queries are machine-translated, and the back-translated variant came from clean English, so 0.805 is an upper bound, not a result for naturally typed Hinglish.

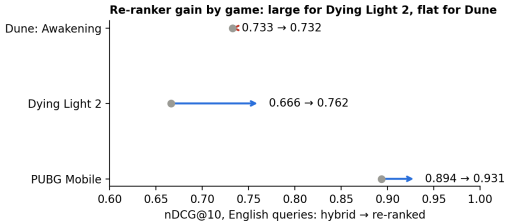
Phase 3: Where the Gains Come From

By query form, and by game



nDCG@10 by query form and stage (223 queries)

- **Game routing (BM25):** naming-based routing lifts 0.636 to 0.677 ($p < 0.001$); guessing unnamed games hurts recall.



Re-ranker gain by game (English queries)

- **Dune is flat after re-ranking** (0.733 → 0.732): 80% of its relevant documents exceed the 512-token limit.

Phase 3: Research Gap and Expected Contribution

Hybrid retrieval is established; our contribution is how its parts interact for gaming queries

Research gap: How do lexical retrieval, multilingual dense retrieval, game-aware routing and re-ranking interact for gaming queries with specialized terminology and different query forms (English, Devanagari, Romanized)?

What We Have Established

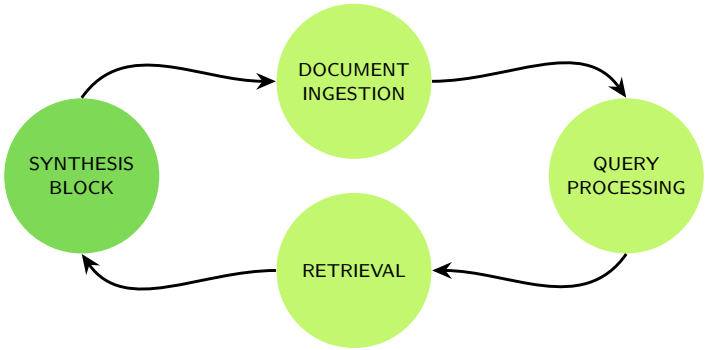
- **Stage-by-stage ablation:** each component's gain measured and significance-tested.
- **Filter placement:** filtering after fusion keeps Dune's ranking intact (0.773 vs. 0.733).
- **Routing variants:** name-based routing beats vote-based guessing.
- **Script gap:** measured, then narrowed to 0.805 vs. 0.823 English with query variants and re-ranking.

Next Experiments

- **Lock in the routing gain:** re-rank on the filter-after-fusion lists.
- **Unlock Dune:** passage-level scoring (MaxP; Dai & Callan, 2019) for long patch notes.
- **Real players' Hinglish:** extend from generated to naturally typed, code-mixed queries.
- **Deployment view:** latency and compute per stage.

Expected contribution: If the gains hold across games and query forms, Phase 3 stands alone as a study of multilingual gaming retrieval; either way, the best pipeline becomes the fixed hand-off (top 5 distinct documents) to Phase 4.

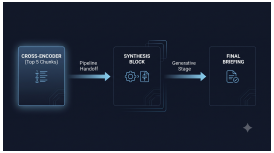
Phase 4: Synthesis and Summarization



Research question: How can retrieved evidence be converted into concise, complete and evidence-grounded answers?

Phase 4: Objective and Baseline

Objective & pipeline hand-off



- **The Goal:** Turn fragmented gaming documents (wiki entries, patch notes, news) into cohesive, fact-dense briefings.
- **The Input:** The top 5 distinct re-ranked documents from Phase 3.
- **The Output:** A structured answer that condenses evidence without unsupported claims.

Baseline Paper: RECOMP (Xu et al., 2024)

Compresses retrieved documents (extractive sentence selection or abstractive multi-document summary) before generation, and compares against plain retrieve-then-read.

Our baseline: Retrieve-then-read on a *fixed* retrieved set and *fixed* context budget; every technique must beat it at the same budget.

Xu, Shi & Choi (2024), *RECOMP: Improving Retrieval-Augmented LMs with Context Compression and Selective Augmentation*. ICLR 2024. Paper
Methodological reference: Laitenberger, Manning & Liu (2025), *Stronger Baselines for Retrieval-Augmented Generation with Long-Context Language Models*. EMNLP 2025. Paper

Experiment 1: Mitigating "Lost in the Middle"

Context ordering, redundancy and compression

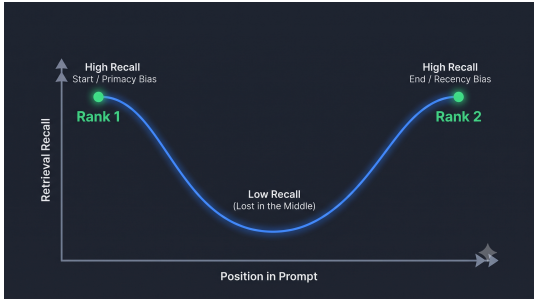
The Challenge:

- LLMs use information in the middle of a long context less reliably than at its start or end (Liu et al., 2023).

Methods, one hypothesis each:

- **LongContextReorder** (evidence position): ranks [1, 3, 5, 4, 2] put key evidence at the prompt boundaries.
- **Redundancy filter** (repeated content): cosine distance removes duplicate staeps.
- **Contextual compression** (context length): extract core facts from mid-ranked documents.

Design: Test each method alone, then combined (filter → compress → reorder); keep only what helps. Same baseline and context budget throughout.



Answer accuracy vs. position of the relevant document (Liu et al., 2023)

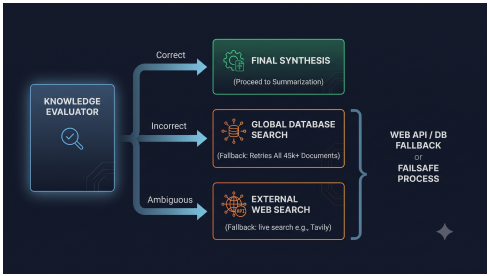
Experiment 2: Corrective Retrieval and Structured Reasoning

Corrective RAG (Yan et al., 2024)

- **Evaluation:** Grades retrieval as Correct, Incorrect or Ambiguous.
- **Fallbacks:** Triggers a wider corpus search or web search when evidence is missing.
- **Hypothesis:** Helps when retrieved evidence is insufficient or unreliable.

Chain-of-Thought and Few-Shot Formatting

- **Hypothesis:** CoT improves multi-step reasoning (Wei et al., 2022); we test whether it also helps reconcile conflicting, version-dependent gaming advice before drafting.
- **Few-shot exemplars:** Enforce a consistent output format.



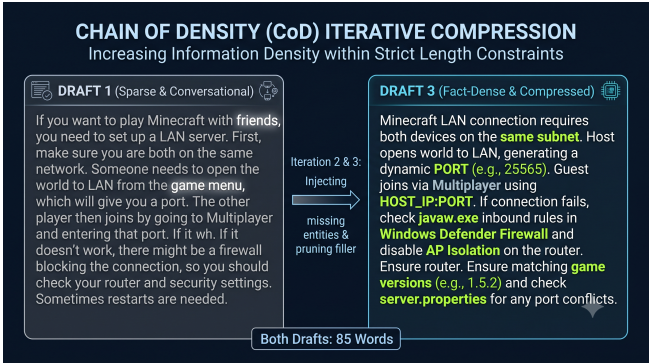
Self-correcting routing tree

Experiment 3: Chain-of-Density Summarization

Iterative densification

- **The Challenge:** Summaries suffer from "lead bias" and miss technical details (Adams et al., 2023).
- **The Mechanism:** A sparse first draft is rewritten twice, each time adding 1–3 missing entities at a fixed length.
- **Comparison:** Normal summary vs. densified summary.
- **Measures:** Factual coverage, unsupported claims, redundancy, completeness and length.

Goal: A more useful, evidence-grounded answer, not just a denser-looking one.



Sparse draft vs. fact-dense draft

Phase 4: Research Gap and Evaluation

Every synthesis method is compared against the simple baseline

Research gap: RECOMP shows compression helps on general open-domain QA. Does compression-style synthesis still help on gaming documents dense with item names and patch versions, and do reordering, correction and densification add measurable gains in coverage and answer quality, under a fixed retrieval set and context budget, without more unsupported claims or unacceptable cost?

Evaluation

Dimension	Question
Answer quality	Correct against the ChronoPlay reference answer?
Evidence support	Is each claim cited and supported by a retrieved document? (ALCE)
Coverage	Are the key facts present?
Cost	Extra LLM calls, tokens and latency?

Decision Rule

- **Standalone:** a method shows a repeatable benefit over the baseline at equal budget.
- **Otherwise:** use the simplest supported synthesis strategy.

Baseline: Xu, Shi & Choi (2024), RECOMP, ICLR 2024 Paper.
Evaluation: Gao et al. (2023), *Enabling Large Language Models to Generate Text with Citations* (ALCE), EMNLP 2023 Paper.

References

Adams et al. (2023). From Sparse to Dense: GPT-4 Summarization with Chain of Density Prompting.

Ammann et al. (2025). Question Decomposition for Retrieval-Augmented Generation. ACL SRW 2025.

Campello et al. (2013). Hierarchical Density Estimates for Data Clustering, Visualization, and Outlier Detection.

Carbonell & Goldstein (1998). The Use of MMR, Diversity-Based Reranking for Reordering Documents and Producing Summaries. SIGIR.

Chari et al. (2025). As cited on the Phase 3 results slide.

Chen et al. (2024). M3-Embedding (BGE-M3). Findings of ACL 2024.

Cormack et al. (2009). Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods. SIGIR.

Dai & Callan (2019). Deeper Text Understanding for IR with Contextual Neural Language Modeling. SIGIR.

Dai et al. (2023). Promptagator: Few-shot Dense Retrieval From 8 Examples. ICLR.

Gala et al. (2023). IndicTrans2.

Gao, Yao & Chen (2021). SimCSE: Simple Contrastive Learning of Sentence Embeddings. EMNLP.

Gao et al. (2023). Enabling Large Language Models to Generate Text with Citations (ALCE). EMNLP.

Grootendorst (2022). BERTopic.

He et al. (2026). ChronoPlay: A Framework for Modeling Dual Dynamics and Authenticity in Game RAG Benchmarks. ICLR.

Karpukhin et al. (2020). Dense Passage Retrieval for Open-Domain Question Answering. EMNLP.

Kulkarni & Callan (2015). Selective Search: Efficient and Effective Search of Large Textual Collections. ACM TOIS.

Laitenberger, Manning & Liu (2025). Stronger Baselines for Retrieval-Augmented Generation with Long-Context Language Models. EMNLP.

Li et al. (2024). DMQR-RAG: Diverse Multi-Query Rewriting for RAG.

Liu et al. (2023). Lost in the Middle: How Language Models Use Long Contexts.

Madhani et al. (2023). Aksharantar: Open Indic-language Transliteration Datasets and Models.

Nogueira & Cho (2019). Passage Re-ranking with BERT.

Rousseeuw (1987). Silhouettes: A Graphical Aid to the Interpretation and Validation of Cluster Analysis. J. Comput. Appl. Math.

Sarthi et al. (2024). RAPTOR: Recursive Abstractive Processing for Tree-Organized Retrieval. ICLR.

Wang, Yang & Wei (2023). Query2doc: Query Expansion with Large Language Models. EMNLP.

Wang et al. (2024). Multilingual E5 Text Embeddings: A Technical Report.

Wei et al. (2022). Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. NeurIPS.

Xu, Shi & Choi (2024). RECOMP: Improving Retrieval-Augmented LMs with Context Compression and Selective Augmentation. ICLR.

Yan et al. (2024). Corrective Retrieval Augmented Generation.

Zheng et al. (2023). Take a Step Back: Evoking Reasoning via Abstraction in Large Language Models.

Zhuang & Zuccon (2022). CharacterBERT and Self-Teaching for Improving the Robustness of Dense Retrievers on Queries with Typos. SIGIR.